



國家實驗研究院
國家高速網路與計算中心
National Center for High-performance Computing

NCHC AI-Powered Physics Bootcamp

March 10-11, 2026

NCHC X OpenACC X NVIDIA

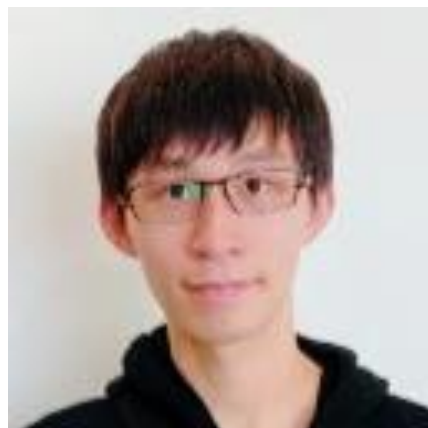
NVIDIA NeMo Bootcamp - Team Roster

Special Thanks to NCHC Team

- 吳宙錦 - 教育訓練系列主辦人
- 郭芳安/潘怡倫/粘美雅 - 晶創雲 計算資源團隊
- 甘恆全博士 - Day 2 邀請分享



Instructor
Chao-Shun Zhan



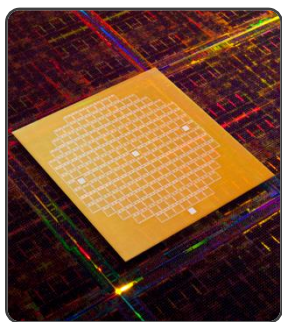
Teaching Assistants
Johnson Sun



Teaching Assistants
Jay Chen

CUDA-X Accelerates Every Industry

6M+ Developers & 900+ SDKs/Models



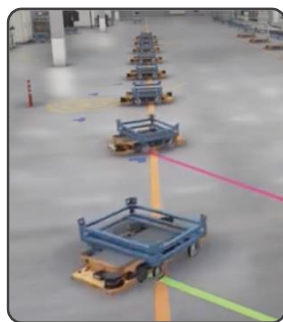
cuLitho

Computational
Lithography



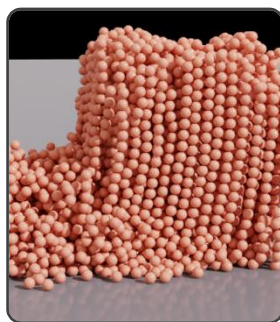
cuDSS

CAE



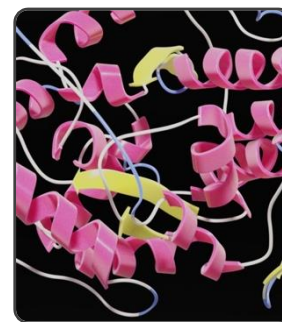
cuOpt

Decision Optimization



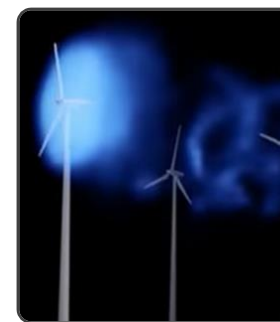
Warp

Physical Simulation



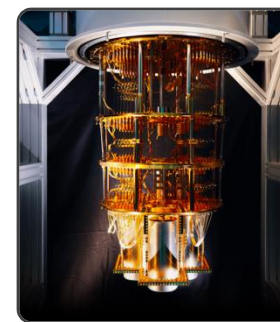
cuDF

Data Processing



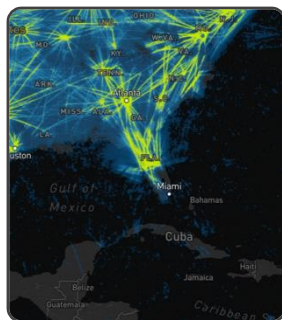
PhysicsNeMo

AI Physics



CUDA-Q

Quantum Computing



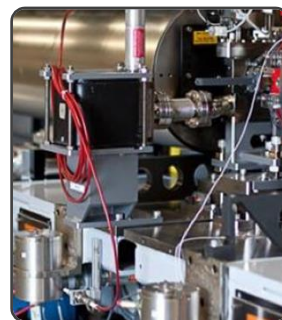
cuEquivariance

Drug & Materials
Discovery



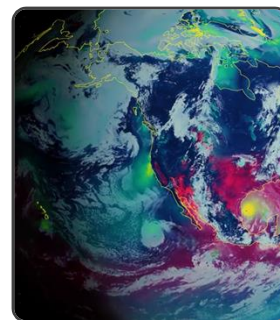
ALCHEMI

AI Materials Science



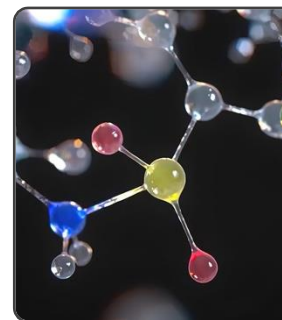
Holoscan

Edge HPC



Earth-2

Weather Analytics



Parabricks

Gene Sequencing



cuPyNumeric

Numerical Computing

AI4Research: A Survey of Artificial Intelligence for Scientific Research

<https://arxiv.org/abs/2507.01903>

by Q. Chen, M. Yang et al.

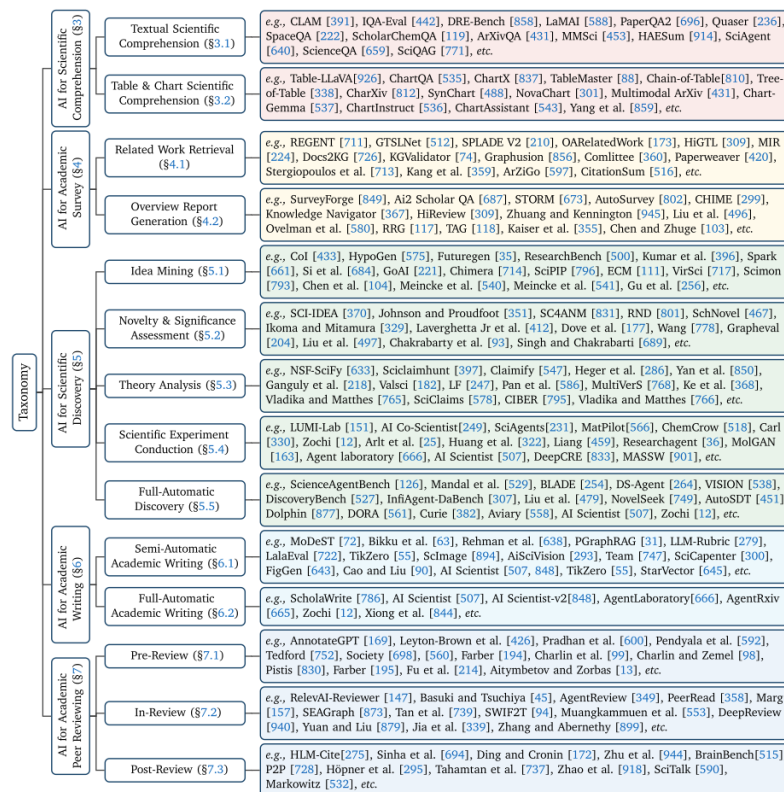
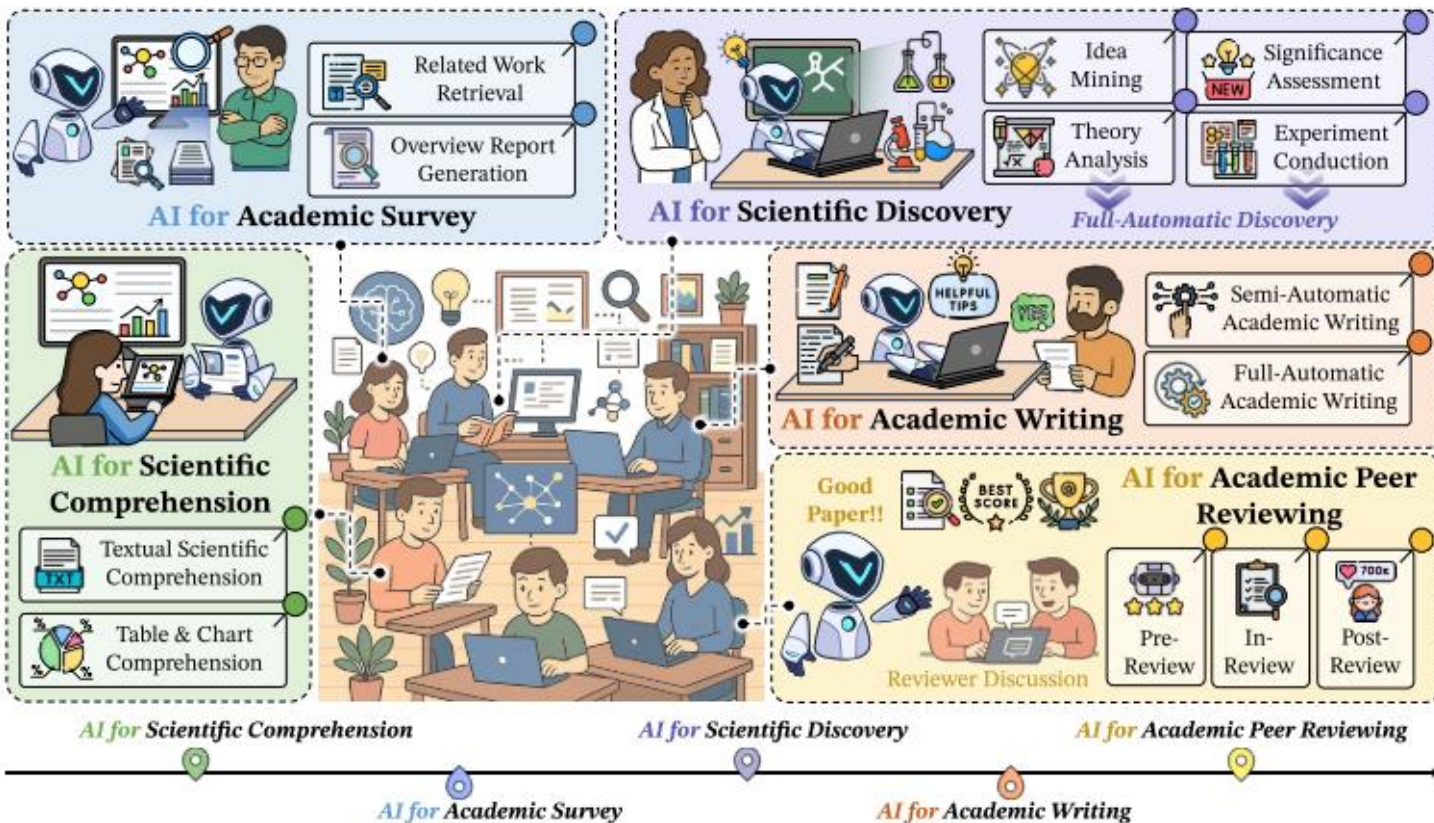


Figure 2: The taxonomy of AI in research (AI4Research) is categorized into five key areas. Each area is subdivided into specific tasks, underscoring the varied roles of AI in the entire research process.

NCHC x NVIDIA Workshops/Bootcamps/Hackathons

Since 2020

Events

50+

Developers

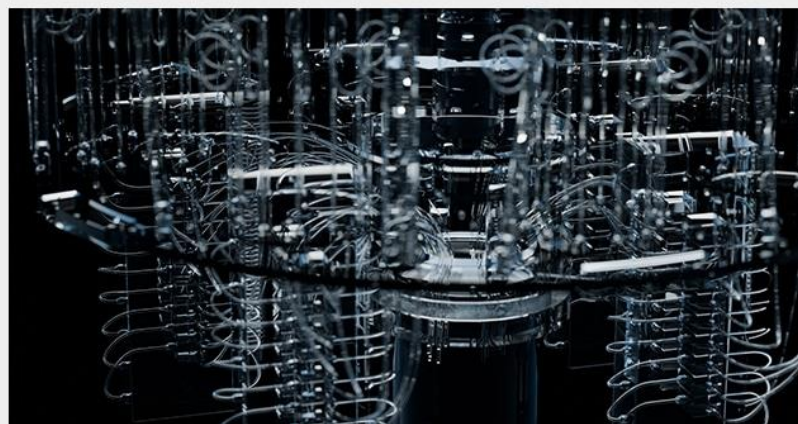
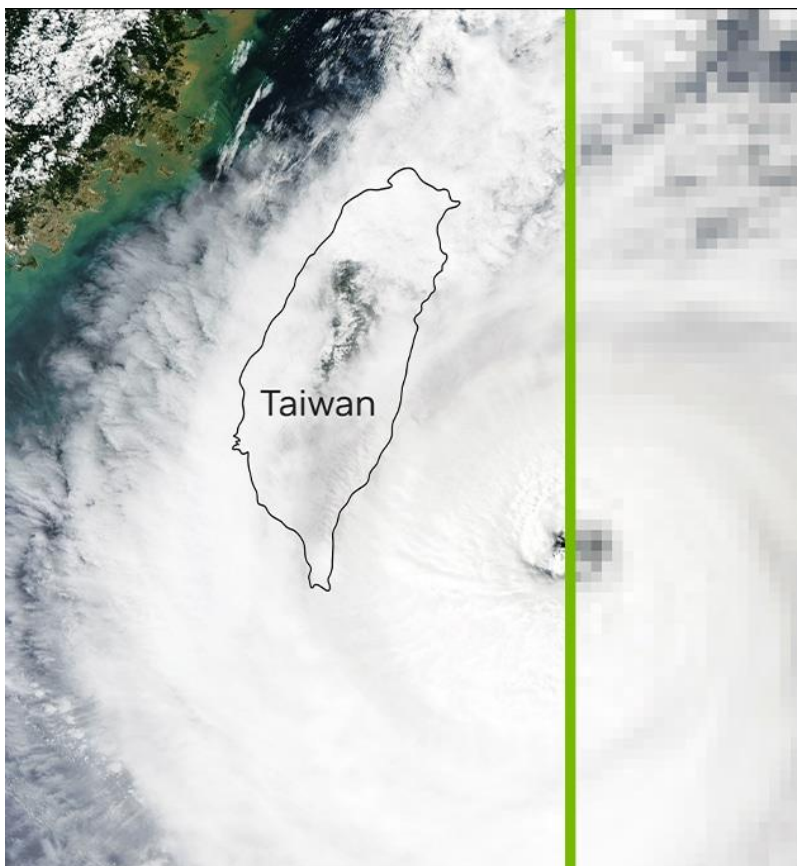
2300+



NCHC 超級電腦為台灣研究帶來大躍進

220*H200-NVL8 + 2*GB200-NVL72

<https://blogs.nvidia.com.tw/blog/taiwan-research-supercomputer/>



國網中心重要超級電腦成長史

<https://www.nchc.org.tw/UploadImage/PageEdit/638920899584150350.pdf>



Co-innovate with Developers

3 Ways

Bootcamps

Training

Open
Hackathon

Acceleration

NVAITC
Projects

Collaboration

Co-innovate with Developers

3 Ways

Bootcamps

Training

Open
Hackathon

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NVAITC
Projects

Collaboration

NCHC-NVIDIA Joint Lab

<https://github.com/nqobu/nvidia/>

- tutorials - NVIDIA online courses/tutorials in AI/HPC
- 20210412 - NVIDIA Techniques Sharing 2021
- 20210706 - NCHC Techniques Sharing 2021
- 20211202 - NVIDIA Techniques Update 2021
- 20211221 - AI+HPC: 利用 NVIDIA Modulus 實踐 PINN 於物理模擬
- 20220415 - NVIDIA Techniques Update 2022
- 20220530 - NCHC-NVIDIA Techniques Sharing 2022
- 20220629 - PINN 與 NVIDIA Modulus 實作訓練營
- 20221111 - Quantum Computing Workshop / 量子計算模擬實作
- 20230413 - NVIDIA Techniques Sharing 2023
- 20230517 - NVIDIA Techniques Briefing: NVIDIA Federated Learning
- 20230525 - AI for Science: NVIDIA Modulus 及 NVIDIA Omniverse 實作
- 20230727 - N-Way to GPU Programming Bootcamp / 多 GPU 程式設計訓練課程
- 20230821 - NVIDIA Techniques Salon 2023: Programming the NVIDIA Superchip
- 20231207 - NCHC Open Hackathon 2023
- 20240410 - NCHC Quantum Computing Bootcamp 2024 - NVIDIA CUDA-Q and cuQuantum
- 20240506 - AI for Science: NVIDIA Modulus, NVIDIA Omniverse, and NVIDIA Earth-2
- 20240508 - NCHC Techniques Sharing 2024
- 20240626 - NCHC AI for Science Bootcamp 2024 - NVIDIA Modulus 物理模擬計算
- 20240806 - NCHC End-to-end LLM Bootcamp 2024 - NVIDIA NeMo 大型語言模型框架
- 20240924 - NCHC N-Way Bootcamp 2024 - NVIDIA GPU 加速運算
- 20241114 - Earth-2 Overview
- 20241129 - NCHC x NTU - NVIDIA BioNeMo Protein Design Workshop 2024
- 20241204 - NCHC Open Hackathons 2024
- 20250218 - NCHC Grace Workshop 2025
- 20250415 - NCHC N-Way Bootcamp 2025 - NVIDIA GPU 加速運算

Co-innovate with Developers

3 Ways

Bootcamps

Training

Open
Hackathon

Acceleration

NVAITC
Projects

Collaboration

2024 Hackathon Publications

- 工商: https://www.ctee.com.tw/news/20250408700939-431204?utm=LINE_share_btn
- 經濟: <https://money.udn.com/money/story/5635/8659767>
- 引新聞: <https://innews.com.tw/223620/>
- Line: <https://today.line.me/tw/v2/article/yzvjnGz>

工商時報

證券 金融 理財 產業 房市 國際 兩

首頁 經濟VIP 經濟彭博 即時 要聞 產業 話

商情 > 產學管顧 >

24參賽團隊黑客松同台競技 三能運算極限

2025.04.08 / 12:29 / 工商時報 文 / 陳又嘉



經濟日報

14:06 川普關稅暫緩90天 海、空運市

經濟日報 > 商情 > 熱門亮點

直擊黑客松競賽！挑戰高效能運算極



引新聞

熱搜 娛樂 社會

首頁 > 工商

直擊黑客松競賽 資源 挑戰高效能

2025/04/08 12:10:05

Facebook 分享

(記者張芸瑄 / 綜合報導) 當 AI 人才，國家高速網路與計算中心 (「NCHC Open Hackathon」) 黑客 AI 技術的實作平台。本屆活動共吸引 24 支隊伍報名，篩選 12 支隊伍晉級決賽，並在 3 週內共同探索運算效能的極限，展現驚人技術成果。

理財

直擊黑客松競賽！參賽團隊借力三方資源挑戰高效能運算極限

商傳媒 更新於 2天前 · 發布於 2天前 · service@sunmedia.tw (商傳媒 SUN MEDIA)

追蹤



2025 Hackathon Publications

<https://techorange.com/2026/01/22/nhc-open-hackathon-report-nvidia/>
<https://www.nchc.org.tw/Message/MessageView/4039?mid=165&page=1>
<https://www.nchc.org.tw/Message/MessageView/4038?mid=165&page=1>
<https://www.nchc.org.tw/Message/MessageView/4037?mid=165&page=1>



最新文章 AI 人工智慧 AI 機器人 智慧製造 新零售 金融科技 資安 主題特展 TO Highlight 展會特輯 全新一週

AI 人工智慧 雲端與運算

國網中心及 NVIDIA、OpenACC 舉辦「2025 NCHC Open Hackathon」黑客松，與台灣卓越研究團隊探索新世代 GPU 技術大幅加速強最新 AI、量子運算等研究的嶄新技術創新

NVIDIA © 2026-01-22



熱門文章

1. Starlink「防封鎖」神話破滅？從伊朗斷網實戰看低軌衛星的三大技術硬傷
2. 模型再強也沒用？Google Gemini 祭出「個人化智慧服務」新功能，決勝點不再只是算力
3. 有錢也買不到的「T-glass」：拆解晶片供應鏈最脆弱的一環，為何連蘋果都束手無策？
4. 新研究繞過 HBM 瓶頸：DeepSeek 如何在算力受限下，於西方之外的 AI 競賽領先美國？



2026.01.14

【2025 NCHC, NVIDIA, OpenACC 黑客松】 - 電腦視覺：無人機，運動科學，影像修復



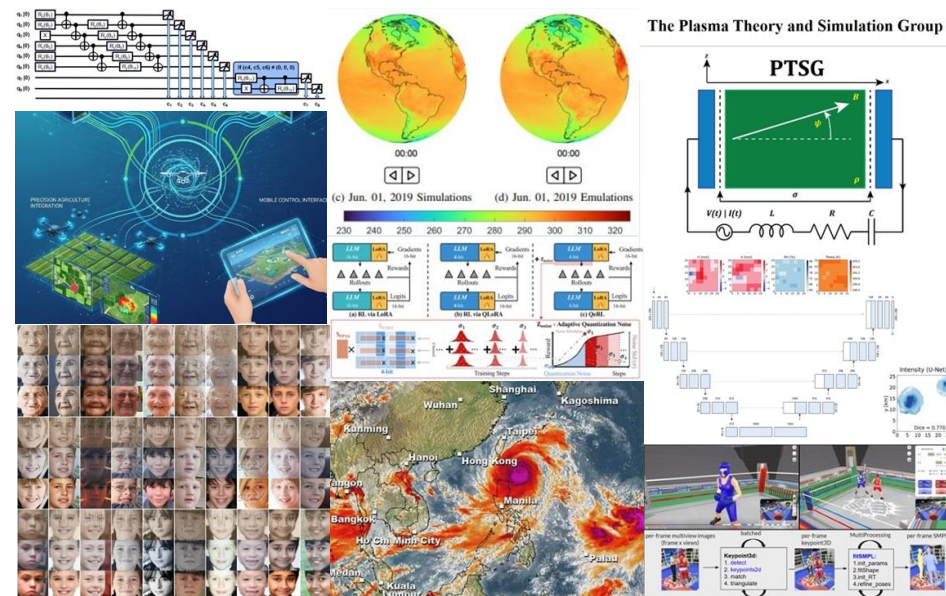
2026.01.14

【2025 NCHC, NVIDIA, OpenACC 黑客松】 - 大氣科學：加速運算，量子算法，人工智慧



2026.01.14

【2025 NCHC, NVIDIA, OpenACC 黑客松】 - 前瞻創新：語言模型，量子化學，電漿模擬



Co-innovate with Developers

3 Ways

CUDA-X
Bootcamp

Training

Open
Hackathon

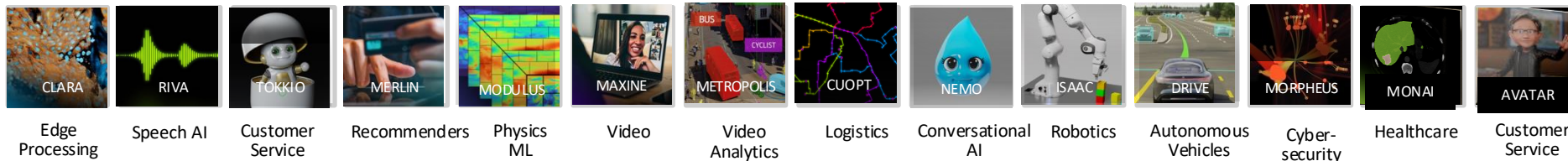
Acceleration

NVAITC
Projects

Collaboration

Platforms for Discovery

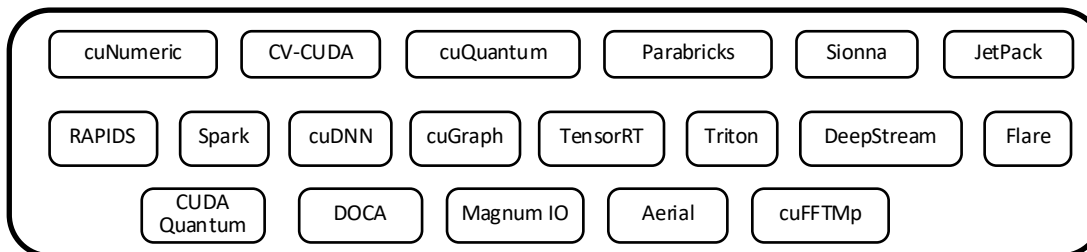
AI APPLICATION
FRAMEWORK



PLATFORMS



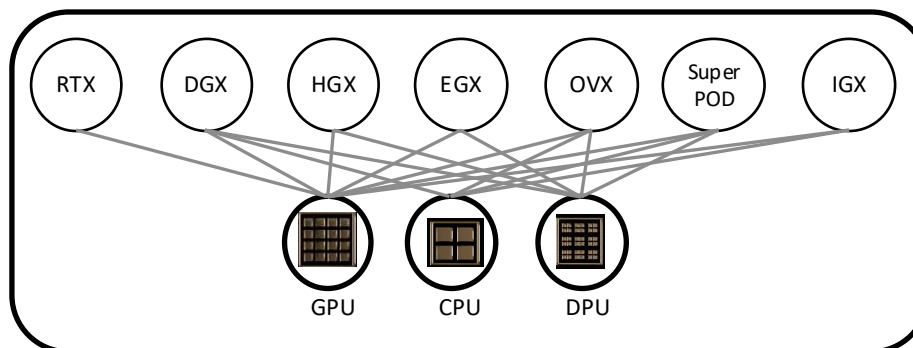
ACCELERATION
LIBRARIES



6M+ Developers
900+ SDKs/Models

CLOUD-TO-EDGE
DATACENTER-TO-ROBOTIC SYSTEMS

3 CHIPS





NVIDIA 開發者計畫

Tools

- 900+ exclusive SDKs and models
- GPU-optimized software, model scripts, and containerized apps
- Early access programs

Training

- Research papers, technical documentation, webinars, blogs, and news
- Technical training and certification opportunities
- 1,000s of technical sessions from industry events On-Demand

Community

- NVIDIA developer forums
- Exclusive meetups, hackathons, and events

Special Program (Present to Jan 2024)

- Join NVIDIA Developer program now, you will get one NVIDIA Training

Join the Community





Reminders for Day 2

NCHC AI-Powered Physics Bootcamp

Day 2 Hands-on

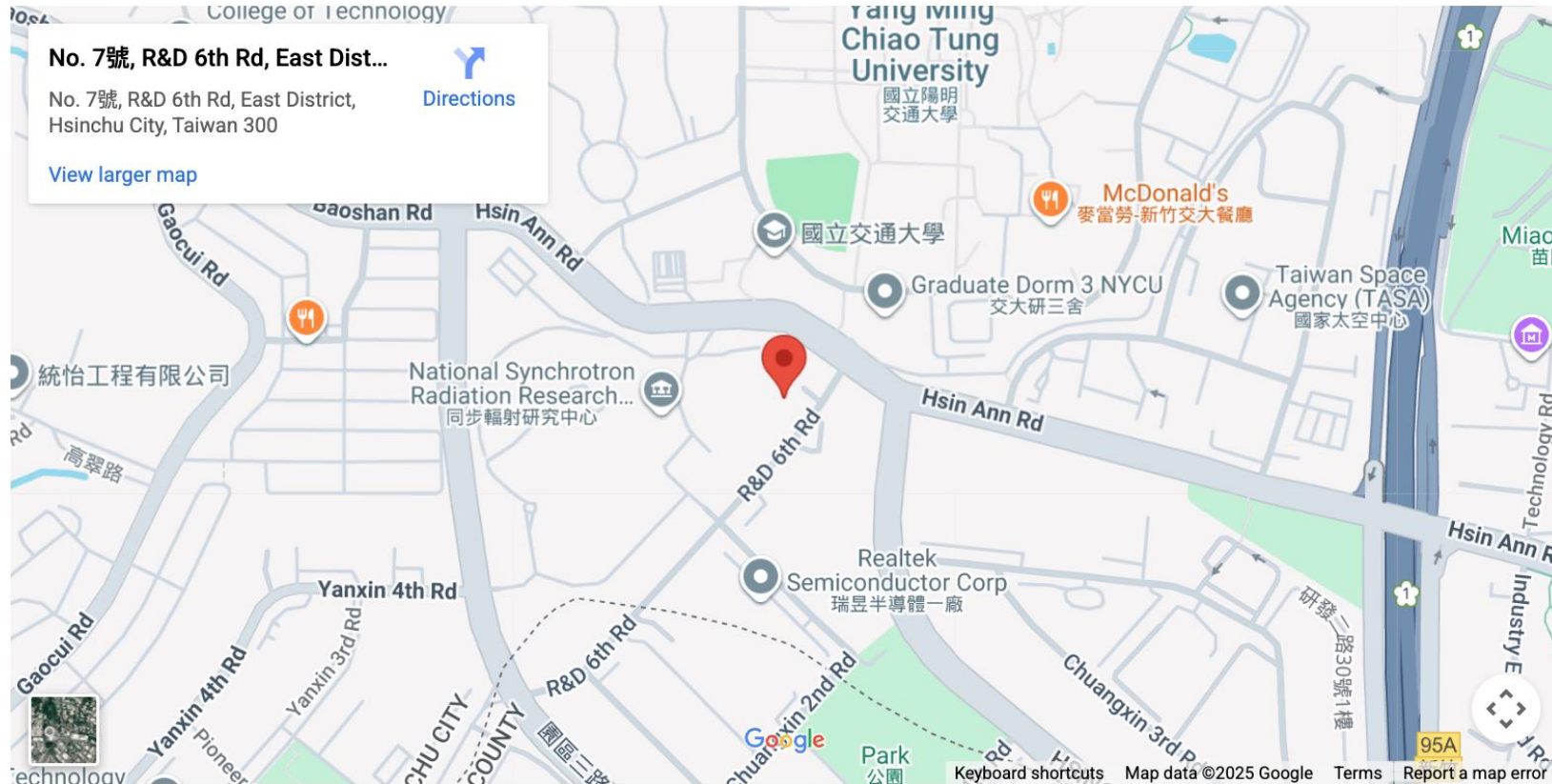
Time	Title
09:30~10:30 (60min)	Recap + Earth2 overview
10:30~11:00 (30min)	DoMINO Overview
11:00~12:00 (60min)	Environment setting
12:00~13:00 (60min)	Break
13:00~16:00 (3hr)	Hands-on : Challenge
13:00~13:45 (45min)	Challenge1: Advanced Wave Dynamics
13:45~14:20 (45min)	Challenge2: Fluid-Structure Interaction
14:20~15:05 (45min)	Challenge3: Multi-Physics Climate Modeling
15:05~16:00 (45min)	Challenge4: Advanced Neural Operators

Transportation to NCHC Hsinchu Office

<https://www.nchc.org.tw/Page?itemid=85&mid=100>

Address: No. 7, R&D 6th Rd., East Dist., Hsinchu City, Taiwan, R.O.C. 30076

Hsinchu traffic during rush hour can be quite heavy. Please allow extra travel time and ensure you arrive by **9:45 AM**.



Parking

Click the button and say hi, the administrator will open the gate.





Day 1 Ending

Experience NVIDIA CEO Jensen Huang's Keynote

Hear from NVIDIA CEO Jensen Huang live on stage on March 17, 3 a.m. (Taiwan Time Zone) at the SAP Center to explore the next chapter in AI.

Arrive early to catch the *GTC Live* 2026 pregame show, where industry leaders will break down the latest AI advances shaping the year ahead.

[Register now](#) before passes sell out.

Workshops March 15

Keynote March 16

Conference and Expo March 16–19

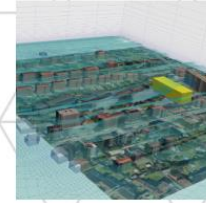
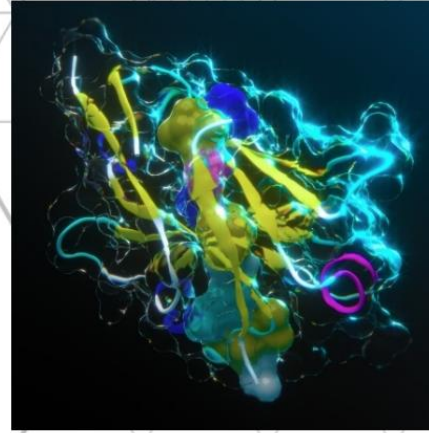


1000+ Talks and Sessions (Register to view online)
Jensen Keynote (more than 43M views last year)

AI for Science Conference Sessions

Learn how accelerated computing is transforming how we model systems, analyze datasets, and perform simulations across scientific disciplines.

[View All AI for Science Sessions](#)



Researcher and Educator Conference Sessions

Discover how the global academic community is transforming science, education, and innovation with the power of AI and accelerated computing.

[View All Researcher and Educator Sessions](#)



Reference

- [Bootcamp Env Setup Instructions] https://github.com/kevin5826536/physicsnemo_bootcamp_setup/tree/main
- [Bootcamp GitHub] <https://github.com/openhackathons-org/AI-Powered-Physics-Bootcamp/tree/main>
- [NCHC 晶創雲 platform] <https://ai-cloud.iic.nchc.org.tw/>
- [NCHC 晶創雲 document] <https://docs.central.iic.nchc.org.tw/user-panel/>
- [PhysicsNeMo GitHub] <https://github.com/NVIDIA/physicsnemo/tree/main>
- [NCHC NVIDIA Joint Lab Events] <https://github.com/nqobu/nvidia/tree/main>
- [NVIDIA GTC 2026] <https://www.nvidia.com/gtc/?ncid=GTC-NVD5I4RZ>
 - 1000+ Talks and Sessions (Register to view online)
 - Jensen Keynote (March 17, 3 AM, > 43M views last year)

Feedback Survey

<https://forms.office.com/r/MM4QFFaPR8>

NCHC AI-Powered Physics Bootcamp Feedback Survey





Thank You